LUIZ F. G. DOS SANTOS, PH.D.

Postdoctoral Associate

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in https://bit.ly/35KSiCn Publications: https://bit.ly/2TV4aik

RESEARCH EXPERIENCE

PostDoctoral Associate at CIRES/University of Colorado Boulder

Using Machine learning for Solar Wind predictions

May 2021 – Current Boulder, CO, USA

- The goal is to use deep learning models (CNNs, GANs, and others) to develop a predictive models of several solar wind quantities.
- Develop a uncertainty quantitative model of the machine learning developed during the process.

Research Assistant at NASA, Goddard Space Flight Center

Developing Machine learning techniques to understand the ICMEs Internal Magnetic field Structure

Jan 2017 – Current Greenbelt, MD, USA

- The goal is to identify patterns (magnetic structures) in time series data and data stream associated with the Sun's ejected events and establish the signatures that validate such structures. We examine the expected structures signatures by synthesizing simple time series data.
- We also use handwriting recognition neural network with synthesized time series data, generated from physical models, to evaluate such structure in magnetic field time sires. The trained network is then used to evaluate observed data from Wind spacecraft.

Analysis of Earth-directed Coronal Mass Ejections

Jan 2017 – Current Greenbelt, MD, USA

- Multi-view data analysis of events based on previous catalogs.
- Analysis of initial stages of Coronal Mass Ejections using coronagraphs (LASCO and COR1&2) observations and tracking the changes in the features throughout the heliosphere.
- Developing a new semi-automated methodology to compare STEREOs images is providing interesting results and showing the implications of the assumptions of radial propagation and self-similar expansion.
- Check the validity of assumptions commonly made for CMEs

Researcher at Frontier Development Lab

Expanding the capabilities of NASA's Solar Dynamics Observatory

Jun 2019 – Current Mountain View, CA, USA

- This project establishes the framework to calibrate extreme ultra-violet (EUV) images and advance our understanding of the cross-relation between different EUV images. We use convolutional neural network (CNN) model that exploits morphological features and solar structures' brightness across single or multiple channels of EUV images.
- We exploit multi-channel data from the Atmospheric Imaging Assembly, onboard the Solar Dynamics Observatory, to demonstrate that unsupervised deep learning can be successfully used to synthetically reproduce images of the solar corona via image-to-image translation and ultimately reduce the telemetry needs of similar space missions.
- Deep learning for deep-space heliophysics missions, including satellite auto-calibration using CNNs, virtual telescopes using image-to-image translation via U-Nets, learned scientific embeddings via GANs, and more.

SKILLS

Python | R | Matlab | PyTorch | Keras | Git | Gitlab | Github | SQL | Google Cloud Computing | CUDA | Numpy | Pandas | Numba | Scikit-Learn | Rapids | XGBoost | SciPy | Data Science | Data Analysis | SunPy | Machine Learning | Computer Vision | Image Recognition | AstroPy | IDL | Fortran | Data Visualization

Hard-working | Focused | Motivated | Collaborative Work | Determined

LANGUAGES

Portuguese ●●●●●
English ●●●●●
Spanish ●●●○○

EDUCATION

Ph.D.in Physics

The Catholic University of America

Sep 2016 – Apr 2021 Washington, DC - USA

- Graduated with distinction
- **GPA: 3.617/4.0**

Master of Science in Physics

The Catholic University of America

Sep 2017 – May 2018 Washington, DC - USA

- **GPA: 3.57/4.0**

Bachelor in Physics

University of Brasília

Sep 2010 – Jul 2015 Brasília, DF - Brazil

- Merit Scholarship granted by the Brazilian Federal Government and the Institute of International Education at Saint John's University
- **GPA: 3.92/4.0**

PUBLICATIONS

Multi-Channel Auto-Calibration for the Atmospheric Imaging Assembly using Machine Learning

A&A Journal

Apr 2021

Luiz F. G. dos Santos, Souvik Bose, Valentina Salvatelli, Brad Neuberg, Mark Cheung, Miho Janvier, Yarin Gal, Meng Jin, Paul Boerner, Atilim Gunes Baydin, *A&A Journal*, Volume 648, Issue A53, Page 12. DOI: https://doi.org/10.1051/0004-6361/202040051

Comprehensive study of an Unusual Widespread Solar Energetic Particle Event on 2013 August 19

A&A Journal

Submitted Nov 2020

Laura Rodríguez-García, Raúl Gómez-Herrero, Yannis Zouganelis, Laura Balmaceda, Teresa Nieves-Chinchilla, Nina Dresing, Mateva Dumbovic, Nariaki Nitta, Fernando M. Carcaboso, Luiz F. G. dos Santos, Lian Jian, L. Mays, David Williams, and Martin J. Rodríguez-Pacheco, *A&A Journal*, Submitted in Nov 2020.

Identifying Flux Rope Signatures Using a Deep Neural Network

Solar Physics Journal

Oct 2020

Luiz F. G. dos Santos, Ayris Narock, Teresa Nieves-Chinchilla, Marlon Nuñez, Michael Kirk, *Solar Physics*, Volume 295, Issue 10, Page 1. DOI: https://doi.org/10.1007/s11207-020-01697-x

Auto-Calibration of Remote Sensing Solar Telescopes with Deep Learning

NeurIPS 2019

Dez 2019

Brad Neuberg, Souvik Bose, Valentina Salvatelli, Luiz F G dos Santos, Mark Cheung, Miho Janvier, Atilim Gunes Baydin, Yarin Gal, Meng Jin. *arXiv*, preprint arXiv:1911.04008

Using U-Nets to Create High-Fidelity Virtual Observations of the Solar Corona

NeurIPS 2019

Dez 2019

Valentina Salvatelli, Souvik Bose, Brad Neuberg, Luiz F G dos Santos, Mark Cheung, Miho Janvier, Atilim Gunes Baydin, Yarin Gal, Meng Jin. *arXiv*, preprint arXiv:1911.04006

Unraveling the Internal Magnetic Field Structure of the Earth-directed Interplanetary Coronal Mass Ejections During 1995–2015

Solar Physics

Feb 2019

Teresa Nieves-Chinchilla, Lan K. Jian, Laura Balmaceda, Angelos Vourlidas, Luiz F. G. dos Santos, Adam Szabo *Solar Physics*, Volume 294, Issue 7, Page 89. DOI: https://doi.org/10.1007/s11207-019-1477-8

ACADEMIC HONORS

2020 John C. Mather Nobel Scholar - July/2020
provided by *John C. Mather and the National Space Grant Foundation* to students that will be recognized as John Mather Nobel Scholars and receive a $\$3,000$ travel allowance towards the cost of presenting research papers at professional conferences.

Google Credits Program Award - Dez/2019
provided by *Google Cloud* to help develop, make cutting-edge discoveries and advance AI research projects providing access to computing power and tools.

Award of Merit - Ago/2019
provided by *NASA and SETI Institute's* at *NASA's Frontier Development Lab* to acknowledge outstanding contribution in the field of artificial intelligence for space exploration and humanity.

Honorable mention - Jun/2015
at *21° UNB Scientific Initiation Congress* for the work done in the project "Transport in turbulent plasma" at University of Brasília

Institutional Scientific Initiation Scholarship Program - Ago/2014
at *University of Brasilia* for the project "Transport in turbulent plasma" at the University of Brasília. This program focus on the education of new researchers under the guidance of a university professor, awarded by the Brazilian National Council for Scientific and Technological Development (CNPq)

Science Without Borders Scholarship 2013-2014
This large scale nationwide scholarship aimed to strengthen and expand the initiatives of science and technology. Study English, be an intern and take a whole year of physics courses at the Saint John's University, awarded by CAPES - Brazilian Coordination for the Improvement of Higher Education Personnel

Constraining the Masses and the Non-radial Drag Coefficient of a Solar Coronal Mass Ejection

The Astrophysical Journal Letters

Mar 2015

C. Kay, L. F. G. Santos and M. Opher *The Astrophysical Journal Letters*, Volume 801, Issue 2, article id. L21, 6 pp. DOI: https://doi.org/10.1088/2041-8205/801/2/L21